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Unscented Transformation Based Robust Kalman Filter and Its Applications in Fermentation Process*

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Abstract State estimation is the precondition and foundation of a bioprocess monitoring and optimal control. However, there are many difficulties in dealing with a non-linear system, such as the instability of process, un-modeled dynamics, parameter sensitivity, *etc*. This paper discusses the principles and characteristics of three different approaches, extended Kalman filters, strong tracking filters and unscented transformation based Kalman filters. By introducing the unscented transformation method and a sub-optimal fading factor to correct the prediction error covariance, an improved Kalman filter, unscented transformation based robust Kalman filter, is proposed. The performance of the algorithm is compared with the strong tracking filter and unscented transformation based Kalman filter and illustrated in a typical case study for glutathione fermentation process. The results show that the proposed algorithm presents better accuracy and stability on the state estimation in numerical calculations.
Keywords robust Kalman filter, unscented transformation, fermentation process, nonlinear system

1 INTRODUCTION

In bioprocess industries, one of the main goals is to increase the yield of the desired product while reducing the energy consumption and production costs. It is important to measure the bioprocess parameters continuously and rapidly for control and on-line optimization. However, appropriate techniques are currently lacking for accurate on-line measurement of process parameters, such as biomass, product concentrations and metabolic rate [1, 2]. Soft-sensor is a useful technology, which has been successfully used to estimate the un-measurable variables in bioprocesses [3, 4] and overcomes the drawbacks of manual analysis and some traditional on-line analytic instrument. State estimation approaches, such as Kalman filter (KF), play an important role in soft-sensor technology, but there are many problems in dealing with fermentation processes, since they are very complicated [5–8].

(1) There are many complex physical and chemical reactions and lots of process parameters.

(2) A fermentation process includes two interested stages, cellular metabolism and extraction of desired products, in which cellular the growth and product synthesis are usually asynchronous.

(3) It is difficult to obtain the yield of desired products from the material balance.

(4) Many factors, such as raw material and cellular quality, substrate components, instruments and operation condition, may affect the fermentation efficiency and make the production unstable.

Some problems affect the performance of KF, such as non-linearity and uncertainties of the model, uncertainties from measurements and different noise sources in the system. The fermentation process is usually described by a nonlinear system. The extended Kalman filters (EKF) can be used for state estimation of nonlinear system, in which linearization is applied to approximate it as a linear time-varying system. In the EKF, calculation of the Jacobian matrices for a high order system is a heavy job for on-line measurement. Further, the linear approximation of the system at a given instant may introduce errors in the mean and covariance of estimated states. Unscented Kalman filters (UKF) [9, 10] avoid the linearization in the EKF update formula by an unscented transformation (UT) and overcome the drawbacks of the EKF. The UKF propagates the probability distribution function (PDF) in a simple and effective way and it is accurate to the second order in estimating the mean and covariance. On the other hand, in most cases, the EKF only gives a biased state estimation, presenting poor robustness against model mismatch. In the UKF, all the sigma points are propagated through the nonlinear system, so the model mismatch also restrains the performance. The strong tracking filter (STF), as proposed by Zhou and Frank [11], introduces sub-optimal fading factor into the prediction error covariance equation of the EKF. The STF has many good characteristics: lower sensitivity to the statistics of initial states and the statistics of the system and/or measurement noise, stronger tracking ability to suddenly changing states no matter whether the filter operates in dynamic or stationary mode, and acceptable computational complexity. In this study, we propose a new algorithm by introducing a sub-optimal fading factor into the UKF and illustrate its performance in a case study.

2 PRELIMINARIES

KF is a recursive state estimator based on the optimal linear filtering theory [12, 13]. The estimation optimum criterion is to minimize the mean-square error.

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$$\mathrm{E}\left\{\left[\boldsymbol{x}(k)-\hat{\boldsymbol{x}}(k)\right]\left[\boldsymbol{x}(k)-\hat{\boldsymbol{x}}(k)\right]^{\mathrm{T}}\right\}\rightarrow\min \tag{1}$$

With this principle, predictive output error sequence (innovation sequence) satisfies the follow condition

$$\mathrm{E}\left\{\tilde{\boldsymbol{y}}(k)\tilde{\boldsymbol{y}}^{\mathrm{T}}(\kappa)\right\}=\boldsymbol{0},\quad 1\leqslant\kappa<k \tag{2}$$

This means that the innovation sequence is not self-correlative.

2.1 EKF principle and algorithm

In the KF algorithm, the computation equations of Kalman gain and covariance matrix are only associated with the state transition matrix and the error transition matrix. To apply KF in a nonlinear system, linearization is an option.

Consider a nonlinear dynamic system equation described as follows:

$$\begin{cases}\boldsymbol{x}(k+1)=\boldsymbol{f}[\boldsymbol{x}(k),\boldsymbol{u}(k),\boldsymbol{w}(k+1),k]\\ \boldsymbol{y}(k)=\boldsymbol{h}[\boldsymbol{x}(k),\boldsymbol{v}(k),k]\end{cases} \tag{3}$$

where $\boldsymbol{x}\in\mathbb{R}^n$ denotes state vector, $\boldsymbol{y}\in\mathbb{R}^m$ denotes output vector, $\boldsymbol{u}\in\mathbb{R}^l$ denotes control vector, $\boldsymbol{w}\in\mathbb{R}^p$ denotes system noise vector, and $\boldsymbol{v}\in\mathbb{R}^q$ denotes observation noise vector. The random vectors $\boldsymbol{w}$, $\boldsymbol{v}$ and $\boldsymbol{x}$ are independent, and their covariance is $\mathrm{E}\{\tilde{\boldsymbol{x}}(k)\tilde{\boldsymbol{x}}^{\mathrm{T}}(k)\}=\boldsymbol{P}(k)$, $\mathrm{E}\{\boldsymbol{w}(k)\boldsymbol{w}^{\mathrm{T}}(k)\}=\boldsymbol{P}_w(k)$, and $\mathrm{E}\{\boldsymbol{v}(k)\boldsymbol{v}^{\mathrm{T}}(k)\}=\boldsymbol{P}_v(k)$, respectively.

The nonlinear system can be transformed to a one-step linear system by using the first-order expression of the Taylor expansion.

$$\begin{cases}\boldsymbol{x}(k+1)=\boldsymbol{F}_x(k)\boldsymbol{x}(k)+\boldsymbol{F}_w(k)\boldsymbol{w}(k+1)\\ \boldsymbol{y}(k)=\boldsymbol{H}_x(k)\boldsymbol{x}(k)+\boldsymbol{H}_v(k)\boldsymbol{v}(k)\end{cases} \tag{4}$$

where

$$\boldsymbol{F}_x(k)=\left.\frac{\partial\boldsymbol{f}(\cdot)}{\partial\boldsymbol{x}^{\mathrm{T}}}\right|_{\substack{\boldsymbol{x}=\hat{\boldsymbol{x}}(k|k)\\ \boldsymbol{u}=\boldsymbol{u}(k)\\ \boldsymbol{w}=\boldsymbol{0}}},\ \boldsymbol{F}_w(k)=\left.\frac{\partial\boldsymbol{f}(\cdot)}{\partial\boldsymbol{w}^{\mathrm{T}}}\right|_{\substack{\boldsymbol{x}=\hat{\boldsymbol{x}}(k|k)\\ \boldsymbol{u}=\boldsymbol{u}(k)\\ \boldsymbol{w}=\boldsymbol{0}}},$$

$$\boldsymbol{H}_x(k)=\left.\frac{\partial\boldsymbol{h}(\cdot)}{\partial\boldsymbol{x}^{\mathrm{T}}}\right|_{\substack{\boldsymbol{x}=\hat{\boldsymbol{x}}(k|k)\\ \boldsymbol{v}=\boldsymbol{0}}},\ \boldsymbol{H}_v(k)=\left.\frac{\partial\boldsymbol{h}(\cdot)}{\partial\boldsymbol{v}^{\mathrm{T}}}\right|_{\substack{\boldsymbol{x}=\hat{\boldsymbol{x}}(k|k)\\ \boldsymbol{v}=\boldsymbol{0}}}.$$

Thus, the KF can be applied to state estimation. This method is the well-known EKF and its algorithm structure is shown in Fig. 1. The EKF algorithm consists of three parts, Riccati equation recursive calculator, Kalman gain calculator and one-step predictor. EKF estimates the state recursively with the initial value $\boldsymbol{x}(\boldsymbol{0})=\boldsymbol{x}_0$ and $\boldsymbol{P}(\boldsymbol{0})=\boldsymbol{P}_0$.

2.2 UKF principle and algorithm

In recent years, an approach named UKF was designed to partially overcome the following drawbacks of traditional EKF and their improved algorithms.

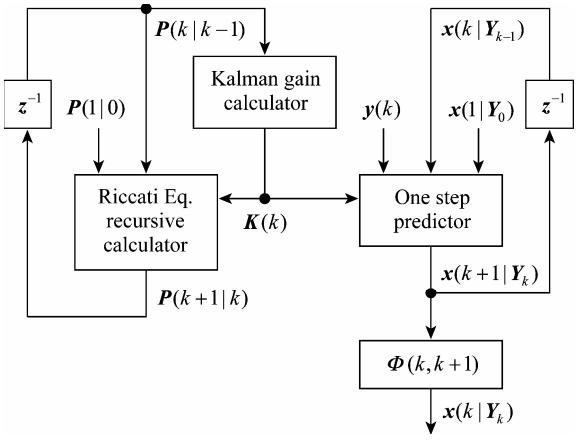


Figure 1 Structural illustration of EKF algorithm

(1) It needs to provide derivatives or Jacobian matrices analytically, so it is difficult to implement, limiting its use in some cases.

(2) The predicted state and the predictive innovation covariance are propagated through two separate linearization systems. This problem may be neglected when the prediction errors are sufficiently small, but this appears rarely.

The main idea underlying the UKF is the use of a propagation technique, termed UT. It uses a minimal set of deterministically chosen sampling points (sigma points) to capture the true mean and covariance of the prior random variables, and when propagated through the nonlinear system, it captures the posterior mean and covariance accurately to the second-order for any nonlinearity without the requirement of the derivatives. Applying UT to KF leads to UKF algorithm, shown in Fig. 2.

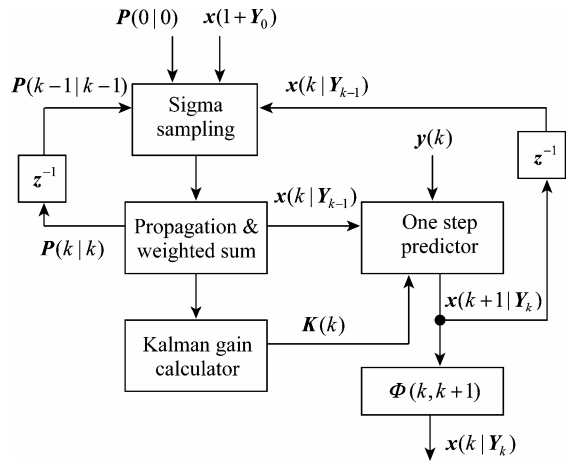


Figure 2 Structural illustration of UKF algorithm

2.3 STF principle and algorithm

EKF works well in some cases, but it can not give an effective estimation in many cases with complex characteristics, such as a bioprocess, due to several reasons.

(1) The system model does not describe an actual process very well because of the modeling method adopted, un-modeled dynamics, system fault, *etc*.

(2) Linearization often leads to unstable filtering performance, and the time-step interval chosen has to be decreased further or the linearized function has to be relinearized when the functions involved are highly nonlinear.

(3) The system is sensitive to initial value of state and covariance.

Usually, the conditional Eq. (2) is not tenable. Some improved KF algorithms have been developed to overcome the above drawbacks. One of them is STF, with its name coming from that (1) it has strong tracking ability to the states whether the states change abruptly or slowly and whether the process reaches steady state or not, and (2) it is fairly robust against model uncertainties.

STF is a robust Kalman filter (RKF), which makes predictive state covariance approaching to zero as much as possible.

$$\mathrm{E}\{\tilde{\boldsymbol{y}}(k)\tilde{\boldsymbol{y}}^{\mathrm{T}}(\kappa)\}\to\boldsymbol{0},\ 1\leqslant\kappa<k \tag{5}$$

Eq. (5) is called "orthogonality principle". STF gives robust estimation by modifying the predictive state covariance with a parameter λ, called fading factor. Contrast with the EKF algorithm, there is one more part in STF, which is used to calculate the factor λ, as shown in Fig. 3. It is calculated as follows:

$$\lambda(k)=1+\frac{\mathrm{trace}\{\boldsymbol{\Sigma}(k)-\boldsymbol{\Sigma}(k\,|\,k-1,\rho)\}}{\mathrm{trace}\{\boldsymbol{\Sigma}_0(k\,|\,k-1)\}} \tag{6}$$

with

$$\boldsymbol{\Sigma}_0(k\,|\,k-1)=\boldsymbol{H}_{\boldsymbol{x}}(k)\boldsymbol{F}_{\boldsymbol{x}}(k-1)\boldsymbol{P}(k-1+k-1)\cdot\boldsymbol{F}_x^{\mathrm{T}}(k-1)\boldsymbol{H}_{\boldsymbol{x}}^{\mathrm{T}}(k)$$

$$\boldsymbol{\Sigma}_0(k\,|\,k-1,\rho)=\boldsymbol{\Sigma}_0(k\,|\,k-1)+\boldsymbol{H}_{\boldsymbol{v}}(k)\boldsymbol{P}_{\boldsymbol{v}}(k)\boldsymbol{H}_{\boldsymbol{v}}^{\mathrm{T}}(k)+\rho\boldsymbol{H}_{\boldsymbol{x}}(k)\boldsymbol{F}_{\boldsymbol{w}}(k-1)\boldsymbol{P}_{\boldsymbol{w}}(k)\cdot\boldsymbol{F}_{\boldsymbol{w}}^{\mathrm{T}}(k-1)\boldsymbol{H}_{\boldsymbol{x}}^{\mathrm{T}}(k)$$

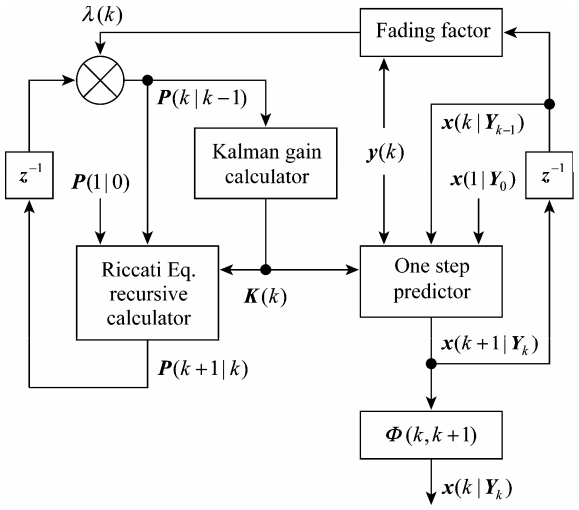


Figure 3 Structural illustration of STF algorithm

It requests $\lambda(k)\geqslant 1$. In Eq. (6), $\boldsymbol{\Sigma}(k)$ denotes actual innovation covariance, $\boldsymbol{\Sigma}(k\,|\,k-1,\rho)$ denotes predictive innovation covariance with compensation, and $\boldsymbol{\Sigma}_0(k\,|\,k-1)$ denotes predictive innovation covariance without noise. The factor ρ is introduced to make the estimation smoother in the numerical calculation and compensate the uncertainty of observation equation.

The subtract matrix of $\boldsymbol{\Sigma}(k)$ and $\boldsymbol{\Sigma}(k\,|\,k-1)$ contains the information of model error, but the actual innovation covariance matrix can not be obtained usually, which is estimated by

$$\boldsymbol{\Sigma}(k)=\mathrm{E}\left\{[\boldsymbol{y}(k)-\hat{\boldsymbol{y}}(k\,|\,k-1)][\boldsymbol{y}(k)-\hat{\boldsymbol{y}}(k\,|\,k-1)]^{\mathrm{T}}\right\} \tag{7}$$

To increase the algorithm stability, the recursive smoothing estimation of $\boldsymbol{\Sigma}(k)$ is introduced.

$$\boldsymbol{V}(k)=\eta\boldsymbol{\Sigma}(k\,|\,k)+(1-\eta)\boldsymbol{V}(k-1) \tag{8}$$

with $\hat{\boldsymbol{y}}(1\,|\,0)=\boldsymbol{h}\left\{\boldsymbol{f}\left[\boldsymbol{x}_0,\boldsymbol{u}(0),\boldsymbol{0}_p,0\right],\boldsymbol{0}_q,1\right\}$ and $\boldsymbol{V}(1)=\mathrm{E}\left\{[\boldsymbol{y}(1)-\hat{\boldsymbol{y}}(1\,|\,0)][\boldsymbol{y}(1)-\hat{\boldsymbol{y}}(1\,|\,0)]^{\mathrm{T}}\right\}$, where $\eta\in[0.5,1]$ is the forgetting factor, and $\eta=0.52$ generally. Thus, the fading factor is formulated as

$$\lambda(k)=\max\left\{1,1+\frac{\mathrm{trace}\{\boldsymbol{V}(k)-\boldsymbol{\Sigma}(k\,|\,k-1,\rho)\}}{\mathrm{trace}\{\boldsymbol{\Sigma}_0(k\,|\,k-1)\}}\right\} \tag{9}$$

The parameters λ and ρ compensate the perturbation in system equation and observation equation, making the estimation robust.

STF algorithm compensates the model perturbation error through predictive output covariance and estimation output covariance and is robust in state tracking when model parameter changes. The STF is a one-step recursion algorithm similar to the standard EKF, and its complexity is moderate.

2.4 Discussion

The difference in the principles of state estimation using EKF, UKF and STF is illustrated in Figs. 1−3. The EKF is applied in a nonlinear system by linearization, in which the mean and covariance are propagated linearly. However, the mean and covariance are actually propagated through nonlinear functions. The linearization results in errors and the states estimation accuracy are limited to the first order moment. The UKF approximates the propagation of the PDF through the nonlinearity more accurately compared to the EKF. With a set of sigma points chosen carefully, which capture the true mean and covariance of a given distribution and pass the mean and covariance of estimated states through a nonlinear transformation, UKF estimates the posterior mean and covariance accurately to the order higher than two. However, since the accuracy of estimation is dependent on that of model, the UKF presents poor robustness against the model mismatch. STF satisfies the "orthogonality principle", in which the Kalman gain matrix is on-line adjusted when model/process mismatch exists. It implies

that all the useful information is extracted from the error vectors.

We will construct an improved KF algorithm that combines the advantages of UKF and STF. An example (soft-sensor system of fermentation process) will be used to proof-test it.

3 UT-RKF ALGORITHM

In the original UKF, the model errors are neglected. In this study, we introduce the robust method of STF into UKF and propose a novel RKF algorithm, termed UT-RKF, the structure of which is shown in Fig. 4.

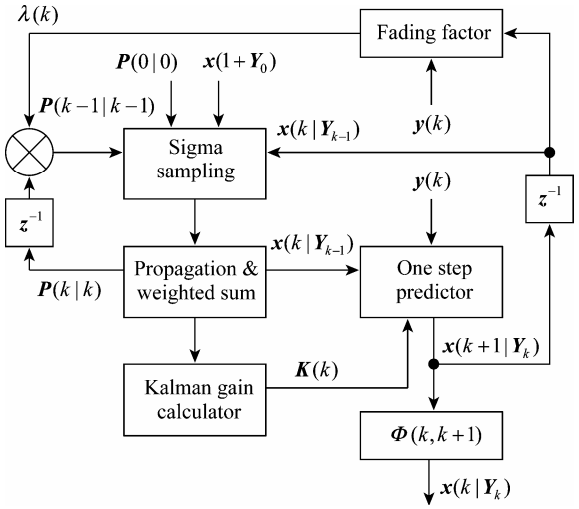


Figure 4 Structural illustration of UT-RKF algorithm

The UT-RKF integrates the advantages of the STF algorithm and the UKF algorithm. It consists of following steps:

(1) Initialization. Construct an extended state $\boldsymbol{z}$ including state $\boldsymbol{x}$, system noise $\boldsymbol{w}$ and observational noise $\boldsymbol{v}$.

$$z(k-1)=\begin{bmatrix}\boldsymbol{x}(k-1)\\ \boldsymbol{w}(k)\\ \boldsymbol{v}(k)\end{bmatrix}$$

Accordingly, its mean and covariance are

$$\hat{\boldsymbol{z}}(k-1|k-1)=\begin{bmatrix}\hat{\boldsymbol{x}}(k-1|k-1)\\ \boldsymbol{0}_p\\ \boldsymbol{0}_q\end{bmatrix},$$

$$\boldsymbol{P}_z(k-1|k-1)=\begin{bmatrix}\boldsymbol{P}(k-1|k-1) & \boldsymbol{0}_{n\times p} & \boldsymbol{0}_{n\times q}\\ \boldsymbol{0}_{p\times n} & \rho\boldsymbol{P}_{\boldsymbol{w}}(k) & \boldsymbol{0}_{p\times q}\\ \boldsymbol{0}_{q\times n} & \boldsymbol{0}_{q\times p} & \boldsymbol{P}_{\boldsymbol{v}}(k)\end{bmatrix}$$

Calculate the corresponding weighting matrices of the first order moments and second order moments. The weight vector $\boldsymbol{\omega}$ and matrix $\boldsymbol{\Omega}$ for extended state $\boldsymbol{z}$ are

$$\boldsymbol{\omega}_{2N+1}=\begin{bmatrix}\dfrac{\gamma}{N+\gamma}\\ \dfrac{\gamma}{2(N+\gamma)}\boldsymbol{1}_{2N}\end{bmatrix},$$

$$\boldsymbol{\Omega}_{2N+1}=\begin{bmatrix}\dfrac{\gamma}{N+\gamma}+\left(1-\alpha^2+\beta\right) & \boldsymbol{0}_{2N}^{\mathrm{T}}\\ \boldsymbol{0}_{2N} & \dfrac{\gamma}{2(N+\gamma)}\boldsymbol{I}\end{bmatrix}$$

where $N=(n+p+q)$. And the weight vector $\boldsymbol{\omega}$ and matrix $\boldsymbol{\Omega}$ for state $\boldsymbol{x}$ are

$$\boldsymbol{\omega}_{2n+1}=\begin{bmatrix}\dfrac{\gamma}{n+\gamma}\\ \dfrac{\gamma}{2(n+\gamma)}\boldsymbol{1}_{2n}\end{bmatrix},$$

$$\boldsymbol{\Omega}_{2n+1}=\begin{bmatrix}\dfrac{\gamma}{n+\gamma}+\left(1-\alpha^2+\beta\right) & \boldsymbol{0}_{2N}^{\mathrm{T}}\\ \boldsymbol{0}_{2n} & \dfrac{\gamma}{2(n+\gamma)}\boldsymbol{I}\end{bmatrix}$$

where $\gamma=\alpha^2(n+\kappa)-n$ is a scaling parameter. The constant α controls the spread of $\boldsymbol{X}$ around the vector $\boldsymbol{x}$ and its range of value is generally $0.001<\alpha<1$. The constant κ is a parameter, which is set to zero for state estimation and $(3-n)$ for parameter estimation. The constant β is related with the prior distribution of $\boldsymbol{x}$, and $\beta=2$ when $\boldsymbol{x}$ obeys the normal distribution.

Set the initial values of the state $\hat{\boldsymbol{x}}(0|0)=\boldsymbol{x}_0$, and the covariance $\boldsymbol{P}(0|0)=\boldsymbol{P}_0$, $\boldsymbol{P}_{\boldsymbol{w}}(0)=\boldsymbol{P}_{\boldsymbol{w}0}$, $\boldsymbol{P}_{\boldsymbol{v}}(0)=\boldsymbol{P}_{\boldsymbol{v}0}$.

(2) Sampling. Calculate sigma points from the given mean and covariance.

$$\boldsymbol{Q}(k-1)=\sqrt{n+\gamma}\,\boldsymbol{P}^{1/2}(k-1|k-1)$$

$$\boldsymbol{X}(k-1)=\hat{\boldsymbol{x}}(k-1|k-1)\boldsymbol{1}_{2n+1}^{\mathrm{T}}+[\boldsymbol{0}_n\quad\boldsymbol{Q}(k-1)\quad-\boldsymbol{Q}(k-1)]$$

$$\boldsymbol{Q}_z(k-1)=\sqrt{N+\gamma}\,\boldsymbol{P}_z^{1/2}(k-1|k-1)$$

$$\boldsymbol{Z}(k-1)=\hat{\boldsymbol{z}}(k-1|k-1)\boldsymbol{1}_{2N+1}^{\mathrm{T}}+[\boldsymbol{0}_N\quad\boldsymbol{Q}_z(k-1)\quad-\boldsymbol{Q}_z(k-1)]$$

The matrix $\boldsymbol{P}_z^{1/2}(k|k)$ is the square root of the $\boldsymbol{z}$ covariance matrix. The transform result is adapted no matter what algorithm is used to solve the square root, so the Cholesky decomposition algorithm can be adopted.

(3) Modification. Calculate the fading factor by standard UT method.

(i) Predict the output covariance without noise.

$$\boldsymbol{Y}(k|k-1)=\boldsymbol{h}\{\boldsymbol{f}[\boldsymbol{X}(k-1),\boldsymbol{u}(k-1),k-1],k\}$$

$$\tilde{\boldsymbol{Y}}(k|k-1)=\boldsymbol{Y}(k|k-1)\left[\boldsymbol{I}-\boldsymbol{\omega}_{2n+1}\boldsymbol{1}_{2n+1}^{\mathrm{T}}\right]$$

$$\boldsymbol{\Sigma}_0(k|k-1)=\tilde{\boldsymbol{Y}}(k|k-1)\boldsymbol{\Omega}_{2n+1}\tilde{\boldsymbol{Y}}^{\mathrm{T}}(k|k-1)$$

(ii) Predict the output covariance with noise.

$$\boldsymbol{Y}(k\,|\,k-1)=\boldsymbol{h}\{\boldsymbol{f}[\boldsymbol{Z}(k-1),\boldsymbol{u}(k-1),k-1],k\}$$

$$\tilde{\boldsymbol{Y}}(k\,|\,k-1)=\boldsymbol{Y}(k\,|\,k-1)\left[\boldsymbol{I}-\boldsymbol{\omega}_{2N+1}\boldsymbol{1}_{2N+1}^{\mathrm{T}}\right]$$

$$\boldsymbol{\Sigma}(k\,|\,k-1,\rho)=\tilde{\boldsymbol{Y}}(k\,|\,k-1)\boldsymbol{\Omega}_{2N+1}\tilde{\boldsymbol{Y}}^{\mathrm{T}}(k\,|\,k-1)$$

(iii) Estimate the actual output covariance.

$$\boldsymbol{V}(k)=\eta\boldsymbol{\Sigma}(k)+(1-\eta)\boldsymbol{V}(k-1)$$

(iv) Calculate the fading factor.

$$\lambda(k)=\max\left\{1,1+\frac{\mathrm{trace}\{\boldsymbol{V}(k)-\boldsymbol{\Sigma}(k\,|\,k-1,\rho)\}}{\mathrm{trace}\{\boldsymbol{\Sigma}_0(k\,|\,k-1)\}}\right\}$$

(v) Modify the covariance of the extended state $\boldsymbol{z}$ and re-sample the sigma points $\boldsymbol{Z}$.

$$\boldsymbol{P}_z(k-1\,|\,k-1)=\begin{bmatrix}\lambda(k)\boldsymbol{P}(k-1\,|\,k-1) & \boldsymbol{0}_{n\times p} & \boldsymbol{0}_{n\times q}\\ \boldsymbol{0}_{p\times n} & \rho\boldsymbol{P}_{\boldsymbol{w}}(k) & \boldsymbol{0}_{p\times q}\\ \boldsymbol{0}_{q\times n} & \boldsymbol{0}_{q\times p} & \boldsymbol{P}_{\boldsymbol{v}}(k)\end{bmatrix}$$

$$\boldsymbol{Q}_z(k-1)=\sqrt{N+\gamma}\,\boldsymbol{P}_z^{1/2}(k-1\,|\,k-1)$$

$$\boldsymbol{Z}(k-1)=\tilde{\boldsymbol{z}}(k-1\,|\,k-1)\boldsymbol{1}_{2N+1}^{\mathrm{T}}+\begin{bmatrix}\boldsymbol{0}_N & \boldsymbol{Q}_z(k-1) & -\boldsymbol{Q}_z(k-1)\end{bmatrix}$$

(4) One-step prediction. Propagate these sigma points through the state and observation equations.

$$\boldsymbol{X}(k\,|\,k-1)=\boldsymbol{f}[\boldsymbol{Z}(k-1),\boldsymbol{u}(k-1),k-1]$$

$$\hat{\boldsymbol{x}}(k\,|\,k-1)=\boldsymbol{X}(k\,|\,k-1)\boldsymbol{\omega}_{2N+1}$$

$$\boldsymbol{Y}(k\,|\,k-1)=\boldsymbol{h}\{\boldsymbol{f}[\boldsymbol{Z}(k-1),\boldsymbol{u}(k-1),k-1],k\}$$

$$\hat{\boldsymbol{y}}(k\,|\,k-1)=\boldsymbol{Y}(k\,|\,k-1)\boldsymbol{\omega}_{2N+1}$$

$$\boldsymbol{P}(k\,|\,k-1)=\tilde{\boldsymbol{X}}(k\,|\,k-1)\boldsymbol{\Omega}_{2N+1}\tilde{\boldsymbol{X}}^{\mathrm{T}}(k\,|\,k-1)$$

$$\boldsymbol{\Sigma}(k\,|\,k-1)=\tilde{\boldsymbol{Y}}(k\,|\,k-1)\boldsymbol{\Omega}_{2N+1}\tilde{\boldsymbol{Y}}^{\mathrm{T}}(k\,|\,k-1)$$

$$\boldsymbol{R}(k\,|\,k-1)=\tilde{\boldsymbol{X}}(k\,|\,k-1)\boldsymbol{\Omega}_{2N+1}\tilde{\boldsymbol{Y}}^{\mathrm{T}}(k\,|\,k-1)$$

with

$$\tilde{\boldsymbol{X}}(k\,|\,k-1)=\boldsymbol{X}(k\,|\,k-1)\left[\boldsymbol{I}-\boldsymbol{\omega}_{2N+1}\boldsymbol{1}_{2N+1}^{\mathrm{T}}\right]$$

$$\tilde{\boldsymbol{Y}}(k\,|\,k-1)=\boldsymbol{Y}(k\,|\,k-1)\left[\boldsymbol{I}-\boldsymbol{\omega}_{2N+1}\boldsymbol{1}_{2N+1}^{\mathrm{T}}\right]$$

(5) Update. Update the predicted moments by the Kalman update equations.

$$\hat{\boldsymbol{x}}(k\,|\,k)=\hat{\boldsymbol{x}}(k\,|\,k-1)-\boldsymbol{K}(k)\left[\boldsymbol{y}(k)-\hat{\boldsymbol{y}}(k\,|\,k-1)\right]$$

$$\boldsymbol{P}(k\,|\,k)=\boldsymbol{P}(k\,|\,k-1)-\boldsymbol{K}(k)\boldsymbol{\Sigma}(k\,|\,k-1)\boldsymbol{K}^{\mathrm{T}}(k)$$

with

$$\boldsymbol{K}(k)=\boldsymbol{R}(k\,|\,k-1)\boldsymbol{\Sigma}^{-1}(k\,|\,k-1)$$

The main characteristics and restrictions of the UT-RKF are summarized as follows:

(1) The derivation of the prediction and update equations is independent of the state and observation equations and the distribution of noise terms. Therefore, no restriction is imposed to the models and the noise statistics, but the needed moments should exist.

(2) The state space model is not affected in any way. The only approximation is to neglect the cumulants in the order higher than two for system noise and observation noise.

(3) Since the extended state includes parameters, the derivatives with respect to the states are not needed. The state space models that contain non-differentiable terms, which usually cause troubles in other methods such as STF, can be applied directly.

4 CASE STUDY: GLUTATHIONE FERMENTATION PROCESS

Glutathione (L-γ-Glutamyl-L-Cysteinyl-Glycine, GSH) is a tripeptide, which is widely distributed in almost all aerobic organisms. Being one of the important physiological active peptides, GSH has wide application potential in medicine, food, athletic sports, *etc*., so the commercial demand for GSH is increased. High levels of GSH content exist in some strains of yeasts (*Saccharomyces cerevisiae* CCRC 21727|ATCC 7754, *S.cerevisiae*), which are the commonly used microorganisms for commercial GSH production by fermentation approach. Generally, the high-density fermentation can reduce the volume of reactor, improve the productivity, and decrease the waste liquid and the cost of separation and purification for the target product.

4.1 GSH fermentation process description

The metabolic process of *S.cerevisiae* yeast can be expressed by the following three chemical calculation equations [14]:

$$\begin{cases}\mathrm{S}+a_1\mathrm{O}_2\xrightarrow{r_1}b_1\mathrm{X}+c_1\mathrm{CO}_2\\ \mathrm{S}\xrightarrow{r_2}b_2\mathrm{X}+c_2\mathrm{CO}_2+d_2\mathrm{P}\\ \mathrm{P}+a_3\mathrm{O}_2\xrightarrow{r_3}b_3\mathrm{X}+c_3\mathrm{CO}_2\end{cases}\tag{10}$$

where X, S and P denote biomass, substrate and product respectively; a_1, b_1, c_1, b_2, c_2, d_2, a_3, b_3 and c_3 are stoichiometric coefficients; r_1, r_2 and r_3 are reaction rates.

The Crabtree effect in the yeast fermentation restrains the cellular growth. In order to decrease glucose concentration in fermentation broth and prolong the fermentation time, the fed-batch fermentation mode is adopted on GSH production, in which specific nutrition solution is injected into the reactor while the products are not extracted during the fermentation.

The widely used growth kinetic model of baker's yeast *S.cerevisiae* is proposed by Sonnleitner and Käppeli [15]. This model can reflect the respiratory

Table 1 Model parameters and initializing value

μ_X/kg·kg^{-1}·h^{-1}	K_X/kg·kg^{-1}	μ_P/kg·kg^{-1}·h^{-1}	K_P/kg·m^{-3}	C_F/kg·m^{-3}	$C_X(0)$/kg·m^{-3}	$C_S(0)$/kg·m^{-3}	$C_P(0)$/kg·m^{-3}	$V_L(0)$/m^3
0.320	0.0250	0.0160	1.00×10^{-4}	500	1.80	28.0	0.00	5.00×10^{-3}

capacity limitation of the cellular metabolism saturation in the fermentation process.

$$\frac{dC_X(t)}{dt} = r_X(t)C_X(t) - \frac{F_S(t)}{V_L(t)}C_X(t) \quad (11)$$

$$\frac{dC_S(t)}{dt} = -r_P(r)C_X(k) + \frac{F_S(t)}{V_L(t)}\left[C_F - C_S(t)\right] \quad (12)$$

$$\frac{dC_P(t)}{dt} = r_P(t)C_X(t) - \frac{F_S(t)}{V_L(t)}C_P(t) \quad (13)$$

$$\frac{dV_L(t)}{dt} = F_S(t) \quad (14)$$

where C_X, C_S and C_P denote the concentrations of biomass, substrate and product in the liquid phase respectively; V_L denotes the volume of liquid phase in fermentor; C_F denotes the substrate concentration of feed; F_S denotes the feed flow rate; $r_X(t) = b_1r_1(t) + b_2r_2(t) + b_3r_3(t)$ denotes the specific growth rate, and $r_P(t) = d_2r_2(t) - r_3(t)$ denotes the overall specific substrate consumption rate. The reaction rate can be described by Monod model, which is the mechanism model of enzymatic kinetics:

$$r_X(t) = \frac{\mu_X}{1 + K_X / C_X(t)}, \quad r_P(t) = \frac{\mu_P}{1 + K_P / C_P(t)}$$

where μ_X is the maximal specific growth rate; μ_P is the maximal substrate uptake rate; K_X and K_P are kinetic constants.

With the dissolved oxygen concentration (C_{DO}), dissolved CO_2 concentration (C_{DC}) and the stirring power (P_S) as the output variables, the observation equations can be obtained using the support vector machine regression method [16]. We write the observation equations as follows

$$\begin{cases} C_{DO}(t) = h_{DO}\left[C_X(t), C_S(t), C_P(t), V_L(t)\right] \\ C_{DC}(t) = h_{DC}\left[C_X(t), C_S(t), C_P(t), V_L(t)\right] \\ P_S(t) = h_{PS}\left[C_X(t), C_S(t), C_P(t), V_L(t)\right] \end{cases} \quad (15)$$

Many parameters in the actual fermentation process are under the automatic closed-loop control. Since the control setting time is very small compared with the parameter variation, these parameters can be taken as constant in the modeling of fermentation process. Table 1 shows the model parameters and initial values.

4.2 Results of UT-RKF

A soft-sensor system based on UT-RKF is established on a laboratory fermentation system for GSH production to on-line estimate the biomass concentration C_X and the product concentration C_P. The UKF, STF and UT-RKF are used to evaluate the performance of the proposed algorithm.

The GSH fermentation process lasts about 50 hours each batch. Because the biomass concentration changes steadily in the first 25 h, we are interested in the later half period, in which the GSH production becomes more obvious. The experimental results in the last 25 hours are shown in Figs. 5 and 6.

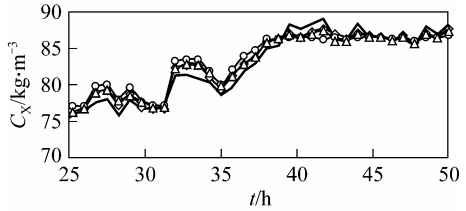


Figure 5 Estimation for biomass concentration
—— experimental value; ◇ UT-RKF estimation; ○ STF estimation; △ UKF estimation

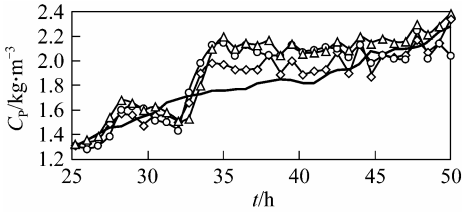


Figure 6 Estimation for product concentration
—— experimental value; ◇ UT-RKF estimation; ○ STF estimation; △ UKF estimation

The biomass growth dynamics is in reactor scale, so the system models describe the dynamics of biomass concentration well. The difference in estimation by the UKF, STF and UT-RKF methods is not distinctive, and the indexes of root-mean-square error (RMSE) are 0.7166, 1.059 and 0.6449, respectively.

The product synthesis dynamics is not only in reactor scale, but also in an organelle scale. The system models established in reactor scale can not predict the dynamics of product concentration accurately. Figure 6 shows that both STF and UT-RKF algorithms track the state change well. The UT-RKF estimation is better then STF due to the application of UT, in which the non-linear system model is directly used to predict the state and covariance, so that estimation accuracy is improved. The RMSE is 0.1422 with UKF estimation, 0.1373 with STF, and 0.0828 with UT-RKF.

The analysis on the experimental results shows that STF and UT-RKF can track the state variables. When the system nonlinearity is diverse, the UT-RKF gives better performance.

5 CONCLUSIONS

This paper presents a UT-RKF algorithm, in which the UT is introduced to improve the accuracy of the mean and covariance of estimated states, and a sub-optimal fading factor is introduced to correct the prediction error covariance. The computation complexity of the UT-RKF algorithm does not increase much. For the representative case study, the present algorithm is slightly better than UKF and STF. Further, the method is implemented in a soft-sensor for predicting the process states on-line.

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